MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **Periodic Test Paper 15 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. If the sum of three numbers in an AP is 9 and their product is 24, the numbers are: [1]

(A) 2, 3, 4 (B) 1, 3, 5 (C) 2, 4, 6 (D) 3, 4, 5

Q2. The 4th term from the end of the AP $-5, -2, 1, \ldots, 49$ is: [1]

(A) 37 (B) 40 (C) 43 (D) 46

Q3. If $a_n = 5n - 3$, then the sum of the first 20 terms is: [1]

(A) 980 (B) 990 (C) 1000 (D) 1020

Q4. If a, b, c are in AP, then the expression $\frac{a-b}{b-c}$ is equal to: [1]

(A) 1 (B) $\frac{a}{c}$ (C) $\frac{b}{a}$ (D) $\frac{c}{a}$

Q5. **Assertion (A):** The sequence $0.2, 0.22, 0.222, 0.2222, \ldots$ does not form an AP.
Reason (R): The differences between consecutive terms $a_2 - a_1 = 0.02$ and $a_3 - a_2 = 0.002$ are not equal. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. Find the sum of all odd integers between 2 and 100 which are divisible by 3. [2]

Q7. Which term of the AP $53, 48, 43, \ldots$ is the first negative term? [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. Divide 56 into four parts in AP such that the ratio of the product of their extremes to the product of their means is 5 : 6. [3]

Q9. If the mth term of an AP is $\frac{1}{n}$ and the nth term is $\frac{1}{m}$, find its $(mn)^{\text{th}}$ term. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. 200 logs are stacked in the following manner: 20 logs in the bottom row, 19 in the next row, 18 in the row next to it, and so on. In how many rows are the 200 logs placed, and how many logs are in the topmost row? [5]

Q11. The sum of four consecutive numbers in an arithmetic progression is 32 and the ratio of the product of the first and the last term to the product of the two middle terms is 7 : 15. Find the four numbers. [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Botanical Garden Flower Bed Planting Rows



In a landscaped flower nursery, rose plants are arranged in tapered decorative rows. There are 23 rose plants in the first row, 21 in the second, 19 in the third, and so on.
The pattern continues uniformly, ending with 5 rose plants in the final row of the flower bed.
Based on the above horticultural arrangement, answer the following questions:

(i) Formulate the AP for the number of rose plants and determine the common difference d. [1]

(ii) How many total rows of rose plants are there in this flower bed? [1]

(iii) Calculate the total number of rose plants planted across the entire flower bed. [3]

OR

(iv) How many rose plants are contained in the 5th row from the front? [3]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★